
A
Intra-Domain Structural Evidence Grounding
Cross-Domain Reasoning Consolidation
Question
Answer
Struct. Def.
Qwen
Reasoning Trace
Structure
Text
SciReasoner Base
Deeper Structure Understanding
Protein-Structure Expert
SFT
RL
- secondary structure
- fold topology
- residue interaction
- active-site geometry
Molecule-Structure Expert
- scaffold topology
- ring system
- bond-disconnection
- stereochemical cue
Material-Structure Expert
- crystal lattice
- unit cell
- defect site
- coordination motif
Self-Bootstrapping
<think>...</think> → Select
Domain-Structure Expert
SciReasoner Base
SFT
RL
SciReasoner ColdStart
SciReasoner
Model
Data
Training
Rollout
Selected Reasoning
Rejected Reasoning
B
All Tasks Dynamics
Structure Tasks Dynamics
Other Task Dynamics
Science Reward
Train Steps (%)
SciReasoner
SciReasoner-Expert
C
pass@1
pass@10
GO avg Fmax (BP/MF/CC)
QMOF avg MAD/MAE
DUD-E EF5%
Retrosynthesis pass@k accuracy
Expert ColdStart
Expert
SciReasoner ColdStart
SciReasoner
GO Term Prediction
QMOF Prediction
3D Molecule Structure (DUD-E)
Retrosynthesis USPTO-50K
D
Expert-ColdStart
Expert-RL
SciReasoner-ColdStart
SciReasoner-RL
Δ mean = +1.79
Δ mean = +1.61
Δ mean = +3.10
Δ mean = +0.48
Density
LLM Judge Score — Overall
E
share of case-judgements (%)
SciReasoner tie or exceed DeepSeek
Overall
GO protein function
Crystalline material
Retro-synthesis
SciReasoner
Strongly prefer
Prefer
Tie
DeepSeek-V4-Pro
Evidence grounding
Domain plausibility
Target-regime alignment
Reasoning coherence
Anti-hallucination
Overall
Human score (1-10)
SciR
DSV4
SciReasoner
DeepSeek-V4-Pro